Denominator Identities for Periplectic Lie Superalgebras

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Background/ Classical Case

Semisimple Lie algebras admit the **Weyl character formula**:

$$\operatorname{ch} L(\lambda) = e^{\rho} \cdot \prod_{\alpha \in \Delta^+} (1 - e^{-\alpha})^{-1} \cdot \sum_{w \in W} (\operatorname{sgn} w) e^{w(\lambda+\rho)}$$

- $L(\lambda)$: simple module
- λ: highest weight
- $\rho = \frac{1}{2} \sum_{\alpha \in \Delta^+} \alpha$
- Δ^+: positive roots
- W: Weyl group

When $\lambda = 0$, $\operatorname{ch} L(\lambda) = 1$ and we obtain the **denominator identity**

$$e^{\rho} \cdot \prod_{\alpha \in \Delta^+} (1 - e^{-\alpha}) = \sum_{w \in W} (\operatorname{sgn} w) e^{w\rho}$$

Examples:

* For $\mathfrak{gl}(n)$, we obtain

$$\prod_{i<j} (x_i - x_j) = \sum_{\sigma \in S_n} (\operatorname{sgn} \sigma)\, \sigma\left(x_1^{n-1} x_2^{n-2} \cdots x_{n-1}\right)$$

* For affine $\widehat{\mathfrak{sl}}(2)$, we get the Jacobi triple product identity.

$$\prod_{m=1}^{\infty} (1 - x^{2m})(1 - x^{2m-1} y^2)\left(1 + \frac{x^{2m-1}}{y^2}\right) = \sum_{n=-\infty}^{\infty} x^{n^2} y^{2n}$$

Denominator Identities for Lie Superalgebras

Representation Theory:

$$1 = \mathrm{ch}\, L(0) = \underbrace{\prod_{\alpha \in \Delta^+} (1-e^{-\alpha})^{-1}}_{R^{-1}} \cdot \sum_{w \in W} (\mathrm{sgn} w)\, e^{w\rho - \rho}$$

$e^{\lambda} \cdot R^{-1}$ is the character of the Verma module $M(\lambda)$ so finding a denominator identity is equivalent to finding $a_\lambda \in \mathbb{C}$ s.t. $\mathrm{ch}\, L(0) = \sum a_\lambda \mathrm{ch}\, M(\lambda)$

For Lie superalgebras the (super)character of the Verma module is

$$\mathrm{sch}\, M(\lambda) = e^{\lambda} \frac{\prod_{\alpha \in \Delta_1^+} (1-e^{-\alpha})}{\prod_{\alpha \in \Delta_0^+} (1-e^{-\alpha})}$$

($\prod_{\alpha \in \Delta_1^+}$: positive odd roots; R_1)

($\prod_{\alpha \in \Delta_0^+}$: positive even roots; R_0)

Def.

$R = \frac{R_0}{R_1}$ is the denominator for Lie superalgebras.

Denominator Identities for Lie Superalgebras

Gorelik-Kac-Wakimoto identities

For basic classical Lie superalgebras

$$e^{\rho} R = \frac{1}{r!} \sum_{w \in W} (\operatorname{sgn} w)\, w\left(\frac{e^{\rho}}{\prod_{\beta \in S} (1 - e^{-\beta})} \right)$$

$r = |S|$

Maximal set of **mutually orthogonal** simple isotropic roots.

Applications:

- studying modules over affine Lie algebras and vertex algebras;
- affine case gives identities in number theory.

Gorelik-Kac-Mösenеder Frajria-Papi

$$e^{\rho} R = \sum_{w \in W} (\operatorname{sgn} w)\, w\left(\frac{e^{\rho}}{(1 - e^{-\beta_1})(1 - e^{-\beta_1 - \beta_2}) \cdots (1 - e^{-\beta_1 - \cdots - \beta_r})} \right), \quad S = \{\beta_1, \ldots, \beta_r\}$$

Application: compact dual pairs.

The Periplectic Lie Superalgebra

$$\mathfrak{p}(n) = \left(\begin{array}{c|c} A & B \\ \hline C & -A^T \end{array}\right) \subset \mathfrak{gl}(n|n)$$

B sym.

C antisym.

$$= \{ A \in \mathfrak{gl}(n|n) \mid (Ax, y) = (-1)^{p(x)p(A)} (y, Ax) \}$$

odd symmtric invariant bilinear form

Even part $\mathfrak{p}(n)_{\bar{0}} \simeq \mathfrak{gl}(n)$

Even roots: $\Delta_{\bar{0}} = \{ \varepsilon_i - \varepsilon_j \mid i \neq j \}$

Odd roots: $\Delta_{-1} = \{ -\varepsilon_i - \varepsilon_j \mid i \neq j \}$, $\Delta_1 = \{ \varepsilon_i + \varepsilon_j \mid i \neq j \} \cup \{ 2\varepsilon_i \}$

Thin denominator: $R = \frac{R_{\bar{0}}}{R_{-1}} = \frac{\prod_{\alpha \in \Delta_{\bar{0}}^+} (1 - e^{-\alpha})}{\prod_{\alpha \in \Delta_{-1}} (1 - e^{-\alpha})}$

Thick denominator: $R = \frac{R_{\bar{0}}}{R_1} = \frac{\prod_{\alpha \in \Delta_{\bar{0}}^+} (1 - e^{-\alpha})}{\prod_{\alpha \in \Delta_1} (1 - e^{-\alpha})}$

Denominator Identities for Periplectic Lie Superalgebras

Theorem (Hoyt-Im-R.)

$$e^{\rho} R = \sum_{w \in W} (\operatorname{sgn} w)\, w\left(\frac{e^{\rho}}{(1-e^{-\beta_1})(1-e^{-\beta_1-\beta_2})\cdots(1-e^{-\beta_1-\cdots-\beta_r})}\right)$$

For the thin denominator, $\beta_1 = -\varepsilon_1-\varepsilon_2$, $\beta_2 = -\varepsilon_3-\varepsilon_4$, ..., $\beta_r = -\varepsilon_{n-1}-\varepsilon_n$ or $-\varepsilon_{n-2}-\varepsilon_{n-1}$

For the thick denominator, $\beta_1 = 2\varepsilon_1$, $\beta_2 = 2\varepsilon_2$, ..., $\beta_r = 2\varepsilon_n$

In the thin case we also have:

Theorem (Hoyt-Im-R.)

$$e^{\rho} \frac{R_0}{R_{-1}} = \frac{1}{r_1} \sum_{w \in W} (\operatorname{sgn} w)\, w\left(\frac{e^{\rho}}{(1-e^{-\beta_1})(1-e^{-\beta_2})\cdots(1-e^{-\beta_r})}\right)$$

Remarks

$e^{-\rho} R^{-1} \cdot \sum_{w \in W} (\operatorname{sgn} w) e^{w(\lambda+\rho)}$ is the supercharacter of thin/thick kac module

Proofs:

Proof 1: (original)

* Hoyt-R: • Evaluation $e^{-\beta_{n-1}-\beta_n}=1$ corresponds to Duflo-Serganova functor.
  • kernel of evaluation is spanned by super characters of Thin Kac modules.

* Action of **translation functors** on thin/thick Kac modules was determined by (Balagoric, Daugherty, Entova-Aizenbud, Halacheva, Hennig, Im, Letzter, Norton, Serganova, Stroppel)

$$1 = ch\,L(0) = (\text{What our formula gives}) + \sum_{\text{finite}} a_\mu\, Kac(\mu)$$

equal after evaluation by induction

in kernel of evaluation

is not annihilated only by one translation functor

is not annihilated by at least two translation functors

Proof 2°

$$e^{\rho} R = \sum_{w \in W} (\operatorname{sgn} w)\, w\,(\text{something})$$

both sides are W-anti-inv.
we calculate orbit representatives

Proof 3° (suggested by Serganova)

$$e^{\rho}\frac{R_0}{R_{-1}} = \sum_{w\in W}(\operatorname{sgn} w)\Big(\sum a_\mu e^{w(\mu+\rho)}\Big) \Longleftrightarrow R_{\pm 1}^{-1} = \sum a_\mu \cdot e^{-\rho} R_0^{-1} \sum_{w\in W}(\operatorname{sgn} w)\, e^{w(\mu+\rho)}$$

the character of the $gl(n)$-module $\operatorname{Sym}(\Lambda^2 V)$ or $\operatorname{Sym}(\operatorname{Sym}^2 V)$ which was studies by Thrall, Howe, Shimura

the character of a simple $gl(n)$-module of highest weight μ